extraterrestrials. (So will some of our present politics, ethics, economics, and religion.) To go beyond present science is one of the chief goals of science. Serious students are not commonly plunged into fits of despair on turning the pages of a textbook and discovering that some further topic is known to the author but not yet to the student. Usually the students struggle a little, acquire the new knowledge, and, following an ancient human tradition, continue to turn the pages.

*"All through history advanced civilizations have ruined civilizations just slightly more backward."* Certainly. But malevolent aliens, should they exist, will not discover our existence from the fact that we listen. The search programs only receive; they do not send.*

THE DEBATE IS, for the moment, moot. We are now, on an unprecedented scale, listening for radio signals from possible other civilizations in the depths of space. Alive today is the first generation of scientists to interrogate the darkness. Conceivably it might also be the last generation before contact is made—and this the last moment before we discover that someone in the darkness is calling out to us.

This quest is called the Search for Extraterrestrial Intelligence (SETI). Let me describe how far we've come.

The first SETI program was carried out by Frank Drake at the National Radio Astronomy Observatory in Greenbank, West Virginia, in 1960. He listened to two nearby Sun-like stars for two weeks at one particular frequency. ("Nearby" is a relative term: The nearest was 12 light-years—70 trillion miles—away.)

Almost at the moment Drake pointed the radio telescope and turned the system on, he picked up a very strong signal. Was it a message from alien beings? Then it went away. If the signal disappears, you can't scrutinize it. You can't see if, because of the Earth's rotation, it moves with the sky. If it's not repeatable, you've learned almost nothing from it—it might be terrestrial radio interference, or a failure of your amplifier or detector … or an alien signal. Unrepeatable data, no matter how illustrious the scientist reporting them, are not worth much.